

Top 50 Python Interview Questions and Answers (Fresher Level)

1. What is Python?

Python is a high-level, interpreted programming language known for its simple and readable syntax.

2. What are the key features of Python?

Easy syntax, interpreted language, dynamic typing, large standard library, and cross-platform support.

3. What are Python data types?

Common data types include int, float, str, list, tuple, set, dict, and bool.

4. What is a variable?

A variable is a name used to store data values in memory.

5. What is a list?

A list is an ordered, mutable collection of elements enclosed in square brackets [].

6. What is a tuple?

A tuple is an ordered, immutable collection of elements enclosed in parentheses ().

7. What is a dictionary?

A dictionary stores data as key-value pairs using curly braces {}.

8. What is a set?

A set is an unordered collection of unique elements.

9. What is the difference between list and tuple?

Lists are mutable while tuples are immutable.

10. What is a function?

A function is a reusable block of code defined using the 'def' keyword.

11. What is a lambda function?

A lambda function is an anonymous function defined using the lambda keyword.

12. What is the difference between == and is?

== compares values while is compares object identity (memory location).

13. What is indentation in Python?

Indentation defines the block of code in Python.

14. What is a module?

A module is a file containing Python code that can be imported and reused.

15. What is a package?

A package is a collection of multiple Python modules.

16. What is exception handling?

It handles runtime errors using try, except, and finally blocks.

17. What is a loop?

A loop executes a block of code repeatedly.

18. What are types of loops in Python?

for loop and while loop.

19. What does break do?

break exits the loop immediately.

20. What does continue do?

continue skips the current iteration and moves to the next one.

21. What does pass do?

pass is a null statement that does nothing.

22. What is a class?

A class is a blueprint for creating objects.

23. What is an object?

An object is an instance of a class.

24. What is inheritance?

Inheritance allows one class to inherit properties from another class.

25. What is polymorphism?

Polymorphism allows the same function name to perform different tasks.

26. What is encapsulation?

Encapsulation hides internal data and restricts direct access.

27. What is abstraction?

Abstraction hides complex implementation details from the user.

28. What is PEP 8?

PEP 8 is the official style guide for writing Python code.

29. What is a constructor?

A constructor is a special method `__init__` used to initialize objects.

30. What is a generator?

A generator is a function that returns an iterator using `yield`.

31. What is list comprehension?

A concise way to create lists using a single line of code.

32. What is slicing?

Slicing extracts a portion of a sequence like a list or string.

33. What is the difference between `append()` and `extend()`?

`append` adds one element while `extend` adds multiple elements.

34. What is the difference between deep copy and shallow copy?

Shallow copy copies references while deep copy copies actual objects.

35. What is a decorator?

A decorator modifies the behavior of a function without changing its code.

36. What is recursion?

Recursion is when a function calls itself.

37. What is `__name__ == '__main__'`?

It ensures that code runs only when the file is executed directly.

38. What is a virtual environment?

It is an isolated environment for Python projects and dependencies.

39. What is pip?

pip is the package manager used to install Python libraries.

40. What is NumPy?

NumPy is a library used for numerical computing with arrays.

41. What is Pandas?

Pandas is a library used for data manipulation and analysis.

42. What is a DataFrame?

A DataFrame is a table-like structure in Pandas.

43. What is Series in Pandas?

A Series is a one-dimensional labeled array.

44. How do you read a CSV file in Python?

Using `pandas.read_csv()`.

45. How do you handle missing values?

Using `dropna()`, `fillna()`, or `replace()` functions.

46. What is data visualization?

It is the graphical representation of data.

47. Which libraries are used for visualization?

Matplotlib, Seaborn, and Plotly.

48. What is EDA?

EDA stands for Exploratory Data Analysis used to understand datasets.

49. What is groupby in Pandas?

groupby is used to group data and perform aggregation operations.

50. What is the difference between merge and join in Pandas?

merge joins data based on columns while join works mainly on index.